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Interval consensus of variable quantity multi-agent systems under complex topology

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ABSTRACT

This paper deals with the complicated tracking problem with variable quantity agents. During the tracking process, the quantity of leader agent set and follower agent set are not fixed, this lead to the dimensions of tracking error and Laplacian dynamic are time-varying. Moreover, every agent imposes boundary condition on consensus value. Specifically, two kinds of interval constraints are introduced which driving system value to desired interval. Then, the consensus analysis becomes more difficult. To solve these issues, multiple Lyapunov functions are constructed to deal with the variable quantity multi-agent tracking process. By the way, we also deal with the finite time consensus with variable quantity agents. Finally, two examples are presented to verify the interval consensus and finite time consensus algorithms.

1. Introduction

Past decades have witnessed the rapid development of multi-agent systems as widely applied in many fields [1–3]. A large number of findings have been presented in various areas, such as consensus under switching topology [4], finite time tracking [5], pinning control [6], disturbance-rejection bounded consensus [7,8], distributed optimization [9–12] and so on [13–15].

There have been increasing works addressing the distributed consensus [7,9,16,17]. In [18], the authors deal with information consensus under dynamical topologies which propose the conclusion that the multiple agents achieve consensus asymptotically only if the union graph contains a spanning tree. Then, second order dynamical systems synchronize are studied in [4], the authors design the controller by sampled position data. The second order system can reach a consensus if the eigenvalues of the Laplacian matrix have at least one nonzero image part. In [19], the stochastic consensus of multi-agent systems is well studied, which just requires the switching topology process can be described by a Markov chain. In [20], the authors presented the second order system could synchronize consensus in finite time. Based on LaSalle's invariance principle, the finite-time synchronization controller is presented by a pinning control algorithm.

Multi-agent system control can be applied to social cluster consensus in [13,16,21,22]. Cluster synchronization for multiple agents are considered in [13], the authors investigate linear multi-agent systems under directed topology with an acyclic graph. In [23], the authors consider the cluster synchronization with nonidentical agents using pinning control. Then, authors in [6] investigate the bounded cluster consensus problem with switching topology which just require the time-varying average topology containing a spanning tree. In [24], the authors solve the cluster synchronization problem with hypergraph. This work achieves synchronization

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and beyond dyadic interactions. In [25], time varying delay is investigated in the distributed control. A Halanay-like inequality is presented to deal with the cluster consensus of delayed coupled agents. In [26], the authors use pinning control to investigate the cluster synchronization problem. However, it is difficult to design multiple asymmetric coupling matrices. Then, the finite time cluster consensus is investigated in [27]. Though using the non-decomposition process, cluster consensus of nonlinear coupled systems is considered.

It is worth mentioning that some researchers consider scaled consensus which is similar with cluster consensus can be found in [28,29,14,30,21,31]. More specifically, the scaled cluster consensus on the structurally balanced graph is studied in [30]. The scaled consensus of heterogeneous system is studied in [29]. A tracking scaled consensus problem under one step time delay is solved by using frequency analysis. In [28], the scaled consensus on switching topology is considered. The authors establish necessary and sufficient conditions to make the coupled systems achieve desired scaled consensus. In [14], the authors investigate a matrix scaled consensus problem, each agent converges to the common point which is inverse by the scaling matrices.

In the tracking process, the external disturbance may change the number of tracking agents.

However, most existing methods deal with the multi-agent system tracking process under switching communication graphs with fixed agent quantity. The dimensions of tracking error and Laplacian dynamic are time invariant. Therefore, these methods are not fit for tracking processes with variable quantity systems. We need to propose a nontrivial method to analyze the tracking error of the variable system set. The previous analysis motivates people to deal with tracking problem with variable quantity of both leader and follower agents.

These papers can be summarized as follows [32–35]. For instance, the authors deal with the multi-agent tracking process with varying follower agents. As the quantity agent set is not fixed, the dimension of tracking errors is jumping. The network topology is also time-varying. The authors propose a nontrivial tracking strategy to overcome the tracking errors under connected undirected topology [35]. Then, the authors extend the results to solve the tracking problem under varying numbers of tracking agent and time delay [32]. A new kind of Voronoi diagram is considered in [34], the authors design a controller to solve the tracking consensus. During the process, the authors present an event-trigger coordination strategy. Different from previous papers, the authors in [33] propose a reinforcement learning method to deal with consensus problem under variable quantity agents.

Most existing papers just consider consensus problem of the varying quantity of followers. However, there are many scenarios that the quantity of leader agent set is time varying [32,35,34,7]. This existing problem motivates us to research the varying number agent set interval consensus under bounded controller in which each agent state is bounded by their own constrain interval or their neighbors' constrain interval.

In this paper, we solve the interval tracking process with variable agent set. The main contributions of this paper can be presented in the following list:

1. We introduce a new tracking scenario which is more practical. The time varying quantity agents interval consensus process of second-order multi-agent systems is considered in this paper, in which the leader set and follower set are quantity varying. The question considered in this paper is an extension of scenario considered in [32,34].
2. In contrast to the methods presented in [32,35,34,7], this paper introduces a simple and rigorous method to deal with the interval consensus under dynamic tracking error, which more practical.
3. A byproduct of our result is that the finite time consensus problem for a series of heterogeneous systems is proved. Compare with the finite-time consensus problem of MASs considered in [7], the problem considered in this paper is more practical, which can be applied to many fields, like, reconnaissance and attack UAV.

The structure of our paper is presented in seven sections. In section 2, we introduce the Notations, some basic definitions about graph theory, interval consensus and system dynamics. Some important Lemmas and definitions are given in section 3. We also analyze the equilibrium point in section 3. Then, Scaled consensus and finite consensus are studied in section 5 and 6. The numerical examples are presented in section 7.

2. Preliminary

2.1. Notations

Let $\mathrm{diag}\{c_1,\dots,c_n\}$ denote a diagonal matrix whose diagonal element are $c_1,\cdots,c_n$. If c is a complex number, $\mathrm{Re}(c)$ and $\mathrm{Im}(c)$ stand for the real and imaginary parts of c, respectively. $M\in\mathbb{R}^{n\times n}$ is a $m\times m$ matrix. $\lambda_1(M)$, $\lambda_2(M)$, $\cdots$, $\lambda_m(M)$ are the eigenvalues of matrix M. con $\{X\}$ denote the convex hull of the set X, that is, $\mathrm{co}\{X\}=\left\{\sum_{j=1}^n\beta_j x_j \mid \sum_{j=1}^n\beta_j=1,\beta_j\geq 0\right\}$; $x_j\in X$, where j and n are positive integers. $\|x\|$ is the Euclidean norm of vector x. $\mathbf{I}_n\in\mathbb{R}^{n\times n}$ stand for a $n\times n$ dimension identity matrix; $\mathbf{0}_n\in\mathbb{R}^{n\times n}$ stand for a $n\times n$ dimension zero matrix; Sometimes, we drop the subscript; let $\mathbf{I}$ and $\mathbf{0}$ stand for compatible dimensions matrices.

2.2. Extended degree

Next, we will introduce the dilation and homogeneity to analyze the finite-time consensus [20]. Let $G(x)$ be K degree homogeneous function, whose dilation are $(s_1,s_2,\cdots,s_n)$, $s_i>0$, if $\forall\epsilon>0$, the following equality is satisfied $F(\epsilon^{s_1}x_1,\epsilon^{s_2}x_2,\cdots,\epsilon^{s_n}x_n)=\epsilon^K F(x)$.

Consider the dynamic system $\dot{x}(t) = G(x)$, if for any $\epsilon > 0$,

$$G_i(\epsilon^{s_1} x_1, \epsilon^{s_2} x_2, \cdots, \epsilon^{s_n} x_n) = \epsilon^{K+s_i} G_i(x),$$

we call the function $G(x) = [G_1(x), G_2(x), \cdots, G_n(x)]^T$ is K degree homogeneous with dilation $(s_1, s_2, \cdots, s_n)$.

2.3. Graph theorem

Consider a series of interconnected agents where the communication topology can be modeled as $G = \{V, E, \mathscr{A}\}$, where V is the agent set, $E \subseteq V \times V$ is the edge set; $A = [a_{ij}] \in \mathbb{R}^{n\times n}$ denotes the adjacency matrix whose elements are positive or zero. The elements in A describe the weighted of communication status. If agent i receives message from agent j, $(j,i) \in E$ and $a_{ji} > 0$; In this paper, we assume $a_{ii} = 0$. Let agent j is the neighbor of agent i, we use $N_i = \{j \mid j \in N, (j,i) \in E\}$ to denote the neighbor set of node i. We define the Laplacian matrix $L = [l_{ij}] \in \mathbb{R}^{n\times n}$ of communication topology G as $L = D - A$, where

$$D = \text{diag}\left\{\sum_{i=1}^{n} a_{1j}, \sum_{i=1}^{n} a_{2j}, \ldots, \sum_{i=1}^{n} a_{nj}\right\}.$$

In the tracking process, one or more leader agents are involved in the communication graph. If agent i can get information from leader agent, $b_i > 0$, otherwise $b_i = 0$. We adopt $B = \text{diag}\{b_1, b_2, \ldots, b_N\}$, then, one can get the extended Laplacian matrix as following: $\tilde{L} = L + B$.

Then, consider a communication topology which is consisted of N agents. Denote $F = \{1, 2, \ldots, M\}$ is the follower set; $E = \{M+1, M+2, \ldots, N\}$ is the leader set.

The communication topology is modeled by graph G. The Laplacian matrix is L, which is in the following form:

$$\begin{bmatrix} L_1 & L_2 \\ 0_{M\times(M-N)} & L_{(M-N)\times(M-N)} \end{bmatrix}$$

2.4. System dynamics

In this subsection, we present the system dynamics and interested problem. This paper investigates the tracking problem, with the following second-order MAS whose evolution dynamics is

$$\begin{cases} \dot{\mathbf{x}}_i(t) = \mathbf{v}_i(t) \\ \dot{\mathbf{v}}_i(t) = -k v_i(t) + u_i \end{cases} \tag{1}$$

where $x_i \in \mathbb{R}$ is system position; $v_i \in \mathbb{R}$ is system velocity; $k > 0$ is a positive constant which stands for damping gain; u_i is the controller.

In our paper, we mainly deal the leader dynamics with following form:

$$\begin{cases} \dot{\mathbf{x}}_i(t) = \mathbf{v}_i(t) \\ \dot{\mathbf{v}}_i(t) = -k v_i(t) \end{cases} \tag{2}$$

where x_i, v_i, k are defined as previous.

Definition 1. Consider the second order system with variable number of tracking agents. Coupled system (1) achieves interval consensus if the following equality satisfies

$$\lim_{t\to\infty} dis(x_i(t), co(x_k(t))) = 0 \tag{3}$$

is satisfied, where $dis(\cdot,\cdot)$ denote the distance in Euclidean norm between vector x_i and set $co(x_k(t))$, $i \in F$, $k \in E$.

Remark 1. From the definition, if the convex hull is just one desired trajectory, the definition of interval consensus becomes classical consensus in [28,29].

Remark 2. We consider the tracking process in this paper, which the quantity of Multi-agent system set is time varying. The tracking process with varying agent set is presented in Fig. 1. Red squares denote leader agents; The green circles denote follower agents. The ① to ⑤ stand for the tracking process stage. Black arrow lines are the evolution flow stage. Furthermore, we describe the sequence of the evolution in Fig. 1. Initially, there are seven agents, then, one leader agent and one follower agent leave: The leader quantity is two; The follower quantity is four, and so on.

As shown in Fig. 1, the interconnected agent set is time-varying, leading to the dimension of communication graph and tracking errors are varying. We assume agent i only gets information from its neighbors, then, we design the controller for tracking agent i as following:

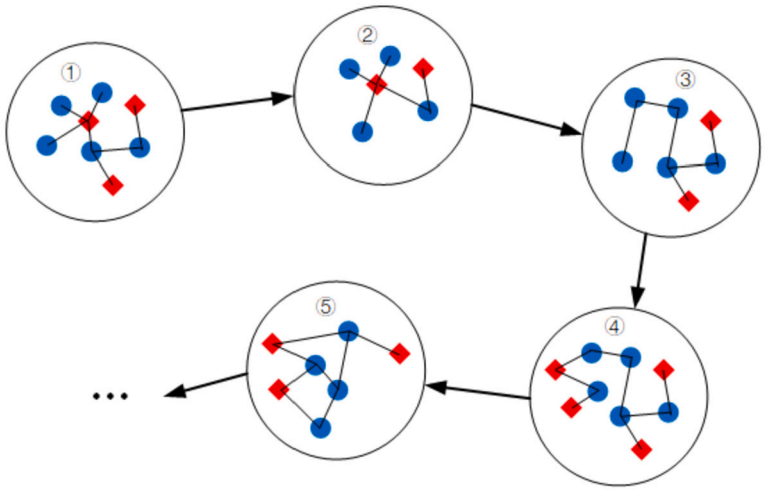


Fig. 1. Illustration of variable number agents. The blue circle represents follower, the red lozenge represents the leader. The leaders and followers have time-varying number.

$$u_i(t) = -\sum_{j\in N_i - E} a_{ij}[H_{ij}(x_j(t)) - x_i(t)] - \sum_{j\in E} a_{ij}[H_{ij}(x_j(t)) - x_i(t))]$$

where E is the time-varying Leader set; N_i is the neighbor agent set of agent i; H_{ij} is a continuous function.

Remark 3. We consider two kinds of controller design: one is agent interval consensus, i.e. $H_{ij} = \mathcal{H}_j$; the other is neighbor interval consensus, i.e. $H_{ij} = \mathcal{H}_i$. Both can be seen as two metrics of social occasion. All people send their own opinion to persuade neighbors adopt the opinion in their interval can be seen as agent interval consensus. If agent just holding their opinion against neighbor's opinion is called neighbor interval consensus, where function $\mathcal{H}_i$ is defined as following

$$\mathcal{H}_i(x) = \begin{cases} a_i, & \text{if } x > a_i \\ x, & \text{if } a_i \le x \le b_i \\ b_i & \text{if } x < b_i \end{cases} \tag{4}$$

where $i = 1, \cdots, N$.

3. Assumptions and lemmas

In this section, we introduce the assumption and some important lemmas.

Assumption 1 *([13]).* We assume the communication topology graph G contains a spanning tree across the time interval $t \in [t_j, t_{j+1})$, and every follower could access at least one target's information, $j = 1, 2, 3, \cdots$.

Lemma 1 *([32]). Assume the dynamics $\dot{x} = F_i(x)$ has m modes. We design the Lyapunov functions as $V_j(x), j = 1, 2, \ldots, m$. The switching moment is defined as $t_0, t_1, \ldots, t_i, \ldots$. If the dynamics switches into m mode in time interval t_i and t_j, Lyapunov function V_m satisfies the following inequality $V_m(x(t_j)) \le V_m(x(t_i))$, where $t_i < t_j$. Then, we say the switching dynamics $\dot{x} = F_i(x), i = 1, 2, \ldots, m$ is stable.*

Lemma 2 *([20]). Suppose $f(x)$ is a continuous function, let $x = 0$ be the stable equilibrium, system $\dot{x} = f(x)$ is K degree homogeneous with dilation $(s_1, s_2, \ldots, s_n)$ If homogeneity degree $K < 0$, then, one can get the equilibrium of system $\dot{x} = f(x)$ is finite-time stable.*

Lemma 3. *Consider a group of dynamic system*

$$\dot{x}_i = f_i(x), \quad f_i(x_0) = 0, \quad x \in \mathbb{R}^{n_i}, \quad i \in \mathcal{T} \tag{5}$$

where $\mathcal{T}$ stands for the switching index set where we assume the set is finite, $\sigma(t) : [0, \infty) \to \mathcal{T}$ be a piecewise constant function. For a fixed constant k, $f_k(x)$ is a continuous function. If switching system $\dot{x} = f_\sigma(x)$ is asymptotically stable, and for any fixed $k \in \mathcal{T}, \dot{x} = f_k(x)$ can achieve finite-time stable, then system $\dot{x} = f_\sigma(x)$ can achieve stable in finite time.

Proof. As the dynamic (5) is asymptotic stable, there exists a positive constant τ such that $|x(t,x_0)|<\epsilon$, $\forall t>\tau+t_0$, $i\in\mathcal{T}$, where $x_i(t,x_0)$ is the trajectory of the dynamics. Let $\tau_i(x_0)$ be the settling time with initial value x_0. Denote $\tau_i(\epsilon)=\max\{\tau(x_0),\|x_0\|<\epsilon\}$ and $\tau^*=\max\{\tau_i(\epsilon)\}$, $i\in\mathcal{T}$. Then, one can get $\tau^*<\tau$, where τ is the dwell time, and $x(t,x_0)=0$, when $t>t_0+\tau^*+\tau$. The proof is complete. □

4. Main results

In the following, we present the interval consensus of time-varying quantity agent set. We mainly deal with the multi-agent interval consensus under $H_{ij}=\mathcal{H}_j$, $i,j=1,\cdots,N$. As for $H_{ij}=\mathcal{X}_i$, the main results are similar with $H_{ij}=\mathcal{X}_j$ case. The controller is defined as following form:

$$\begin{aligned}u_i(t)&=-\sum_{j\in M}a_{ij}[H_{ij}(x_j(t))-x_i(t)]\\&=\sum_{j\in M\setminus\gamma}a_{ij}(x_j(t)-x_i(t))-\sum_{j\in\gamma}a_{ij}(y_j-x_i(t))\end{aligned}$$

where γ is as set which satisfies $H_{ij}(x_j(t))\neq x_j(t)$. We present them in compact form:

$$\begin{aligned}U=\begin{bmatrix}u_1\\\vdots\\u_N\end{bmatrix}&=-\begin{bmatrix}L_1 & L_2\end{bmatrix}\begin{bmatrix}X_F\\X_E\end{bmatrix}\\&\quad-\begin{bmatrix}\sum_{j\in\gamma}a_{1j} & & \\ & \ddots & \\ & & \sum_{j\in\gamma}a_{Nj}\end{bmatrix}\begin{bmatrix}x_1\\\vdots\\x_N\end{bmatrix}+B\\&=-\tilde{L}_1x_F-L_2x_E+B\end{aligned}$$

where $\tilde{L}_1=L_1+\begin{bmatrix}\sum_{j\in\gamma}a_{1j} & & \\ & \ddots & \\ & & \sum_{j\in\gamma}a_{Nj}\end{bmatrix}$, $B=\begin{bmatrix}b_1 & \cdots & b_N\end{bmatrix}^T$, $b_i=\sum_{j\in\gamma}a_{ij}y_j$, $x_F=[x_1,x_2,\cdots,x_N]^T$, $x_E=[x_{N+1},\cdots,x_M]^T$, $L_1\in\mathbf{R}^{N\times N}$, $L_2\in\mathbf{R}^{N\times(M-N)}$.

Remark 4. If the network topology satisfies Assumption 1, eigenvalues of matrix $\tilde{L}_1$ are positive. The detailed proof can be found in [36].

As $x_F-\tilde{L}_1^{-1}L_2x_E=-\tilde{L}_1^{-1}u+\tilde{L}_1^{-1}B$, let $\delta x_i=x_i-\sum_{j\in E}c_{ij}x_j$, $\delta v_i=x_i-\sum_{j\in E}c_{ij}v_j$ where $[c_{ij}]=\tilde{L}_1^{-1}L_2$, one can get

$$\begin{cases}\delta\dot{x}_i=\dot{x}_i-\sum_{j\in E}c_{ij}\dot{x}_j=v_i-\sum_{j\in E}c_{ij}v_j\\\delta\dot{v}_i=-kv_i+u_i+\sum_{j\in E}c_{ij}kv_j=-k\delta v_i+u_i\end{cases}\tag{6}$$

Then, one can get the compact form of coupled system (6)

$$\begin{bmatrix}\delta\dot{x}\\\delta\dot{v}\end{bmatrix}=\begin{bmatrix}0 & I\\-\tilde{L}_1 & -kI\end{bmatrix}\begin{bmatrix}\delta x\\\delta v\end{bmatrix}+\begin{bmatrix}0\\B\end{bmatrix}\tag{7}$$

The candidate Lyapunov function for system (7) can be presented in following form:

$$V=\begin{bmatrix}\delta x\\\delta v\end{bmatrix}^T\begin{bmatrix}P & 0\\0 & Q\end{bmatrix}\begin{bmatrix}\delta x\\\delta v\end{bmatrix}\tag{8}$$

where P and Q are positive definite matrices.

Then, the tracking process could be transformed into stable process associated with a group of tracking error systems.

Theorem 1. *Consider the multi-agent systems (7) with switching topologies across the time interval $[t_k,t_{k+1})$. If the Assumption 1 is satisfied, the variable number systems will achieve bounded consensus if damping gain and dwell time satisfy the following inequality.*

$$k\geq\frac{Im(\mu_i)}{\sqrt{Re(\mu_i)}}$$

and

$$T_k \geq \frac{\rho - \eta}{\omega},$$

where $T_k \triangleq t_{k+1} - t_k$, ρ, η, ω *are positive constants.*

The proof can be found in the following section.

Remark 5. From Theorem 1, one can find the variable number agents achieve bounded consensus under Assumption 1, has nothing with the number of leader agent set.

Remark 6. We explain the equilibrium point in algebraic viewpoint, which is different from [32,35,34,7].

Then, we will analyze the equilibrium point of dynamics

$$\sum_{j\in N_i} a_{ij}[H_{ij}(x_j(t)) - x_i(t)] = 0, \tag{9}$$

which is valuable for practical applications.

Corollary 1. *Let* $[x_1^*, \cdots, x_N^*]$ *be the solution of system (9) with* $H_{ij}(\cdot) = \mathcal{H}_j, i, j = 1, \cdots, N$, $b^* = \max_j b_j$ *and* $a^* = \min_j a_j$. *We have the following conclusion: (1) if* $b^* \leq a^*$, *then* $b^* \leq x_i^* \leq a^*$, *for all i; (2) if* $a^* \leq b^*$, *then* $a^* \leq x_i^* \leq b^*$, *for all i;*

Proof. From previous proof, one can find $\exists\ i, j$ satisfy $x_i^* \neq x_j^*$, then, we will show that $x_i \in \cap_{j\in V} I_j$.

(1) $b^* \leq a^*$, the proof consists of two subsections. We first prove $x_i^* \leq a^*$. By contradiction method, we assume there exists a set Γ, such that $x_j^* > a^*$ for all $j \in V$. Let $x_s^* = \max_{j\in\gamma} x_j^*$, and a node l of neighbor of s satisfies $x_l^* < x_x^*$.

In the following, if $\mathcal{H}_v(x_v^*) \leq x_v^*$, one can get

$$\mathcal{H}_v(x_v^*) - x_s^* \leq x_v^* - x_s^* \leq 0$$

There exists l neighbor of s, satisfy $x_l^* < x_s^*$, and $\mathcal{H}_l(x_l^*) - x_s^* < 0$. Then, consider $\mathcal{H}_v(x_v^*) > x_v^*$, this condition holds only if $\mathcal{H}_v(x_v^*) = b_v$. As $b^* < a^*$, one can get $b_v \leq b^* \leq a^* < x_s^*$. Then, $\mathcal{H}_v(x_v^*) - x_s^* = p_v - x_s^* < 0$.

If there exists a s satisfies $x_s^* = \max_{j\in\gamma} x_j^* > a^*$, we have $\sum_{j\in N_s} a_{sj}(\mathcal{H}_j(x_j^*) - x_s^*) < 0$ which is a contradiction. Then, $x_j^* \leq a^*$ for all $j \in V$.

(2) $b^* \geq a^*$ We will prove $x_i^* \leq p^*, \forall i \in V$. Suppose this condition is not satisfied, there exists $x_j^* > b^*$, we deal with term $\mathcal{H}_v(x_v^*) - x_s^*$ for v the neighbor of s.

Case 1: $\mathcal{H}_v(x_v^*) \leq x_v^*$, that is, $\mathcal{H}_v(x_v^*) - x_s^* \leq x_v^* - x_s^*$, there exists a node v neighbor of s, such that $\mathcal{H}_v(x_v^*) - x_s^* < 0$.

Case 2: $\mathcal{H}_v(x_v^*) > x_v^*$, we have $\mathcal{H}_v(x_v^*) = b_v$. As $b_* \geq a_*$, $\mathcal{H}_v(x_v^*) - x_s^* = b_v - x_s^* \leq b^* - x_s^* < 0$.

Therefore, there exists node s such that $x_k^* = \max_{j\in V} x_j^* > b^*$, such that $\sum_{j\in N_s} \mathcal{H}_j(x_j^*) - x_s^* < 0$. That is a contradiction. The other proof is similar with (1). The proof is completed. □

5. Scaled consensus

In the following, we analyze the scaled consensus model with variable agents. We denote $x_i' = r_i x_i$, $v_i' = r_i v_i$, then, the model can be presented as

$$\begin{cases} \dot{x}_i' = r_i v_i = v_i' \\ \dot{v}_i' = -k r_i v_i + u_i' = -k v_i' + u_i' \end{cases} \tag{10}$$

For notation simplicity, we also define $\delta x_i' = r_i x_i - \sum_{j\in E} c_{ij} r_i x_j$, then we have $\delta \dot{x}_i' = r_i \dot{x}_i - \sum_{j\in E} c_{ij} r_j x_j = \delta v_i'$. The controller u_i' is defined as following:

$$\begin{aligned} u_i'(t) = &- \sum_{j\in N_i - E} a_{ij}[H_{ij}(r_j x_j(t)) - r_i x_i(t)] \\ &- \sum_{j\in E} a_{ij}[H_{ij}(r_j x_j(t)) - r_i x_i(t))] \\ = &- \sum_{j\in N_i - E} a_{ij}[H_{ij}(x_j'(t)) - x_i'(t)] \\ &- \sum_{j\in E} a_{ij}[H_{ij}(x_j'(t)) - x_i'(t))] \end{aligned} \tag{11}$$

We can write the controller u_i in compact form:

$$U' = \begin{bmatrix} L_1 & L_2 \end{bmatrix} \begin{bmatrix} r_1 x_1 \\ \vdots \\ r_N x_N \end{bmatrix} - \begin{bmatrix} \sum\limits_{j\in\gamma} a_{1j} & & \\ & \ddots & \\ & & \sum\limits_{j\in\gamma} a_{Nj} \end{bmatrix} \begin{bmatrix} r_1 x_1 \\ \vdots \\ r_N x_N \end{bmatrix}$$
$$+ \begin{bmatrix} \sum\limits_{j\in\gamma} a_{1j} & & \\ & \ddots & \\ & & \sum\limits_{j\in\gamma} a_{Nj} \end{bmatrix} \begin{bmatrix} r_1 y_1 \\ \vdots \\ r_N y_N \end{bmatrix}$$
$$= \begin{bmatrix} L_1 & L_2 \end{bmatrix} \begin{bmatrix} x'_1 \\ \vdots \\ x'_N \end{bmatrix} - \begin{bmatrix} \sum_{j\in\gamma} a_{1j} & & \\ & \ddots & \\ & & \sum_{j\in\gamma} a_{Nj} \end{bmatrix} \begin{bmatrix} x'_1 \\ \vdots \\ x'_N \end{bmatrix}$$
$$+ \begin{bmatrix} \sum_{j\in\gamma} a_{1j} & & \\ & \ddots & \\ & & \sum_{j\in\gamma} a_{Nj} \end{bmatrix} \begin{bmatrix} r_1 y_1 \\ \vdots \\ r_N y_N \end{bmatrix}$$

or

$$U' = -\tilde{L}_1 x'_F - L_2 x'_E + \begin{bmatrix} \sum\limits_{j\in\gamma} a_{1j} & & \\ & \ddots & \\ & & \sum\limits_{j\in\gamma} a_{Nj} \end{bmatrix} \begin{bmatrix} r_1 y_1 \\ \vdots \\ r_N y_N \end{bmatrix}$$

Through a short calculation, one can get $x'_F + \tilde{L}_1^{-1} L_2 x'_E = -\tilde{L}_1^{-1} u' + \tilde{L}_1^{-1} B'$, $\delta x' = -\tilde{L}_1^{-1} u' + \tilde{L}_1^{-1} B'$ and

$$u' = -\tilde{L}_1 \delta x' + B'.$$

Then, we have the following equalities:

$$\begin{aligned} \delta \dot{v}'_i &= -k v'_i - u'_i - \sum_{i\in E} c_{ij}(-k v'_j) \\ &= -k v'_i + u'_i + k \sum_{i\in E} c_{ij} v'_j \\ &= -k \delta v'_i + u'_i \\ &= -k \delta v'_i - \tilde{L}_1 \delta x'_i + B' \end{aligned}$$

where $B' = [b_1, \cdots, N]$, $b_i = \sum_{j\in\gamma} a_{ij} r_j y_j$, in which every agent has a scaling gain r_i, and controller u_i update based on the consensus protocol:

$$\begin{bmatrix} \delta \dot{x} \\ \delta \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & I \\ -\tilde{L}_1 & -\kappa I \end{bmatrix} \begin{bmatrix} \delta x \\ \delta v \end{bmatrix} + \begin{bmatrix} 0 \\ B' \end{bmatrix} \tag{12}$$

Consider the coupled system (12), we need to solve the equation $det(\lambda I - \mathcal{A}) = 0$ to calculate the eigenvalues of $\mathcal{A} = \begin{bmatrix} 0 & I \\ -\tilde{L}_1 & -\kappa I \end{bmatrix}$. As $det(\lambda I - \mathcal{A}) = det(\lambda I - \begin{bmatrix} 0 & I \\ -\tilde{L}_1 & -\kappa I \end{bmatrix}) = det(\lambda(\lambda + \kappa) I + \tilde{L}_1)$, if the eigenvalues of $det(\lambda I - \mathcal{A})$ are negative, we just need

$$\kappa > \sqrt{\frac{Im(\mu_i)^2}{Re(\mu_i)}}$$

or

$$\kappa > \frac{Im(\mu_i)}{\sqrt{Re(\mu_i)}}$$

where u_i denotes the eigenvalue of matrix $\mathcal{A}$; $Im(\mu_i)$ and $Re(\mu_i)$ are the image part and real part of μ_i, respectively.

Theorem 2. *Consider the coupled system (10), under Assumption 1 and the controller is designed as (11) across time interval* $[t_k, t_{k+1})$, $k = 0, 1, 2, \cdots$, *then, coupled systems can achieve interval consensus if the following condition holds:*

$$\kappa > \frac{Im(\mu_i)}{\sqrt{Re(\mu_i)}}$$

and

$$T_k \geq \frac{\rho - \eta}{\omega}$$

where ρ, η, ω are positive constants.

Proof. We choose the candidate Lyapunov function for tracking process of multi-agent system (7) as

$$V(t) = \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}$$

where P and Q are positive definite matrices.

For $t \in [t_k, t_{k+1})$, one can get following equations:

$$\begin{aligned}
\dot{V}(t) &= \begin{bmatrix} \delta \dot{x}' \\ \delta \dot{v}' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} + \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta \dot{x}' \\ \delta \dot{v}' \end{bmatrix} \\
&= \left\{ \begin{bmatrix} 0 \\ B' \end{bmatrix}^T + \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}^T \begin{bmatrix} 0 & I \\ -\tilde{L}_1 & -kI \end{bmatrix}^T \right\} \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} \\
&\quad + \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \left\{ \begin{bmatrix} 0 & I \\ -\tilde{L}_1 & -kI \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} + \begin{bmatrix} 0 \\ B' \end{bmatrix} \right\} \\
&= \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix}^T \left\{ \begin{bmatrix} 0 & P \\ -Q\tilde{L}_1 & -kQ \end{bmatrix} + \begin{bmatrix} 0 & -\tilde{L}_1 Q \\ P & -kQ \end{bmatrix} \right\} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} \\
&\quad + 2 \begin{bmatrix} 0 \\ B' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} \\
&\leq -\xi V(t) + 2 \begin{bmatrix} 0 \\ B' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix} \begin{bmatrix} \delta x' \\ \delta v' \end{bmatrix} \\
&\triangleq -\xi \|z\|^2 + c_1 \|z\| \\
&\leq -(1-\theta)\xi \|z\|^2 \qquad \forall \|z\| \geq \mu = \frac{c_1}{\theta \xi}
\end{aligned}$$

where ξ is a positive constant [7], $0 < \theta < 1$ is a constant. c_1 is the biggest eigenvalue of matrix $2 \begin{bmatrix} 0 \\ B' \end{bmatrix}^T \begin{bmatrix} P & \\ & Q \end{bmatrix}$.

Then, we can get $\dot{V} \leq -\omega < 0$, where $\omega = min\{(1-\theta)\xi \|z\|^2\}$. The foregoing inequality implies that

$$V(t, x(t)) \leq V(t_0, x(t_0)) - \omega(t - t_0)$$

which show that $V(t, x(t))$ reduce to η within the time interval $[t_0, t_0 + \frac{\rho - \eta}{\omega}]$.

When $t_{k+1} - t_k \geq T_k \triangleq \frac{\rho - \eta}{\omega}$, $k = 1, 2, \cdots$, we have that

$$\|z\| \leq b, \qquad t > t_0 + \frac{\rho - \eta}{\omega}$$

where $\rho = V(t_0, x(t_0))$, $b = \frac{c_1}{\theta \xi} \sqrt{\frac{\lambda_N}{\lambda_1}}$, λ_1 and λ_N is the eigenvalues of matrix $\begin{bmatrix} P & \\ & Q \end{bmatrix}$.

Then, we can get the conclusion that coupled system (10) under controller (11) will achieve interval consensus in bounded time. □

Remark 7. It is worth point that the upper bound of consensus error is bounded by $b = \frac{c_1}{\theta \xi} \sqrt{\frac{\lambda_N}{\lambda_1}}$. That is to say, the smaller $\frac{\lambda_N}{\lambda_1}$, the thinner interval or consensus error.

6. Consensus in finite time

With the development of unmanned aerial vehicles (UAVs), UAVs are applied in the harsh and dangerous environment to replace human. In order to ensure safety and work efficiency, the finite time control of UAVs is studied by many scholars [37,38]. Therefore, we will analyze the finite time consensus of coupled systems in the following. A nontrivial controller is introduced to make the variable number systems achieve consensus in finite time. The second-order system can be written in the following form:

$$\begin{bmatrix} \delta \dot{x}_i' \\ \delta \dot{v}_i' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -k \end{bmatrix} \begin{bmatrix} \delta x_i' \\ \delta v_i' \end{bmatrix} + \begin{bmatrix} 0 \\ u_i \end{bmatrix} \tag{13}$$

To achieve finite time consensus, we design the following controller.

$$u_i = -sig\left[\sum_{j=1}^{M} a_{ij}(x_i' - x_j'))\right]^{\alpha} - sig\left[\sum_{j=1}^{N} a_{ij}(v_i' - v_j')\right]^{\alpha} \tag{14}$$

where $0 < \alpha < 1$ is positive constant; $sig(\cdot)$ is sign function. Then, we can get the following results.

Theorem 3. *Consider the multi-agent systems (13) with finite time consensus controller (14), if Assumption 1 is satisfied, the coupled system will achieve interval consensus in finite time.*

Proof. We choose the candidate Lyapunov function for system (13) as following

$$V(x(t)) = V_1(x(t)) + V_2(x(t))$$

where $V_1 = \frac{1}{2}\sum_{i=1}^{N} \delta v_i^2$, $V_2 = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\int_0^{\delta x_i - \delta x_j} a_{ij} sig(s)^{\alpha_1} ds$.

As we have

$$\begin{aligned}\dot{V}_1 &= \sum_{i=1}^{N} \delta v_i \delta \dot{v}_i \\ &= \sum_{i=1}^{N} \delta v_i \left\{ -k\delta v_i - sig(\sum_{j=1}^{N} x_i - x_j)^{\alpha} - sig(\sum_{j=1}^{N} v_i - v_j)^{\alpha} \right\}\end{aligned}$$

and

$$\dot{V}_2 = \frac{1}{2}\sum_i \sum_j a_{ij} sig(\delta x_i - \delta x_j)^{\alpha}(\delta v_i - \delta v_j),$$

let $V = V_1 + V_2$, then, we have

$$\begin{aligned}\dot{V} =& \dot{V}_1 + \dot{V}_2 \\ =& \sum_{j=1}^{N} \delta v_i \left[-k\delta v_i + \sum_{j}^{N} a_{ij} sig(\delta x_j - \delta x_i)^{\alpha} \right. \\ & \left. + sig(\delta v_j - \delta v_i)^{\alpha} \right] \\ & + \frac{1}{2}\sum_i \sum_j a_{ij} sig(\delta x_i - \delta x_j)^{\alpha}(\delta v_i - \delta v_j) \\ =& -k\sum_{i=1}^{N} \delta v_i \delta v_i + \sum_{i=1}^{N} \delta v_i sig(\delta v_j - \delta v_i)^{\alpha} \\ =& -k\sum_{i=1}^{N} \delta v_i \delta v_i + \frac{1}{2}\sum_{i=1}^{N} a_{ij}(\delta v_i - \delta v_j) sig(\delta v_j - \delta v_i)^{\alpha} \\ \leq& -k\sum_{i=1}^{N} \delta v_i \delta v_i\end{aligned}$$

Consider $\dot{V} = 0$, one can get

$$-k\sum_{i=1}^{N} \delta v_i \delta v_i + \frac{1}{2}\sum_{i=1}^{N} a_{ij}(\delta v_i - \delta v_j) sig(\delta v_j - \delta v_i)^{\alpha} = 0 \tag{15}$$

According to Assumption 1 and equation (15), one can get $\delta v_i = \delta v_j = 0$, $\forall i, j$. Obviously, $\dot{V} = 0$ if and only if $\delta v_i = \delta v_j = 0$, which implies $\delta \dot{v}_i = 0$, $\forall i, j$. Then, we have $\delta \dot{v}_i = -k\delta v_i + \left[\sum_{j=1}^{M} a_{ij} sig(x_i' - x_j'))\right]^{\alpha} - \left[\sum_{j=1}^{N} a_{ij} sig(v_i' - v_j')\right]^{\alpha} = 0$.

It follows

$$\sum_{i=1}^{N} \delta x_i \left[\sum_{j=1}^{M} a_{ij} sig(x_i' - x_j'))\right]^{\alpha} \tag{16}$$

$$= \frac{1}{2}\sum_{i}^{N}\sum_{j=1}^{M} a_{ij}(\delta v_i - \delta v_j) sig(x_i' - x_j')) \tag{17}$$

$$= 0 \tag{18}$$

Then, we have $\delta x_i = \delta x_j = 0$, $\forall i, j$. According to the LaSalle's Invariance Principle, $\delta x_i \to 0$, $\delta v_i \to 0$, $\forall i$, as $t \to \infty$.

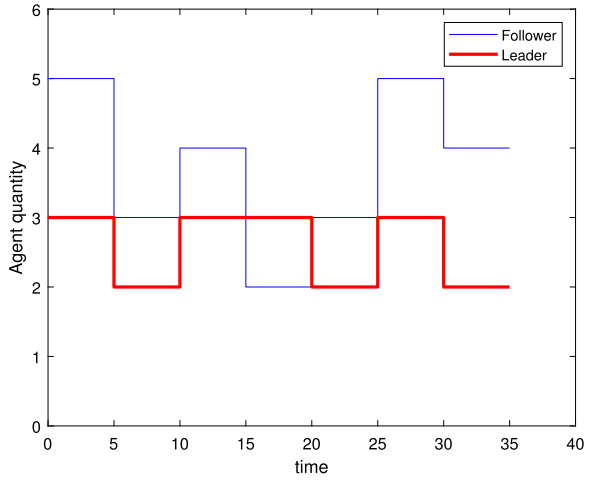


Fig. 2. Illustrative diagram of agent quantity switching. The quantity of leader agents changes from 2 to 3; The quantity of leader agents changes from 2 to 5.

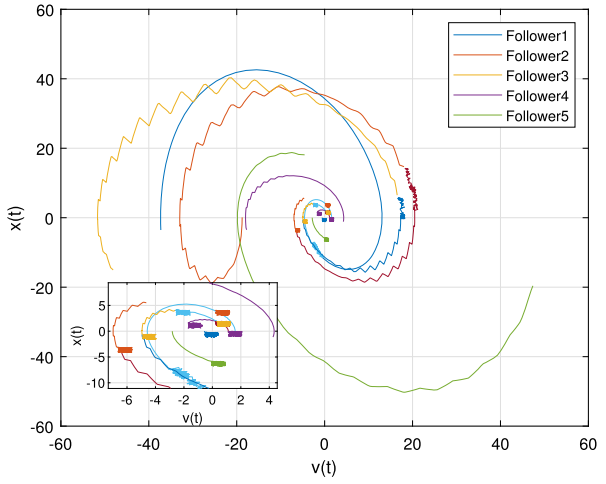


Fig. 3. The positions of follower agents achieve bounded consensus with $k = 2.5$.

Then, one can find the system (13) with variables $(\delta x_1', \delta x_2', \cdots, \delta x_n', \delta v_1', \delta v_2', \cdots, \delta v_n')$ is homogeneous of degree $\alpha - 1$ with dilation $(2, 2, \cdots, 2, 1+\alpha, 1+\alpha, \cdots, 1+\alpha)$.

It follows, system (13) is globally asymptotically stable. System (13) is also locally homogeneous with degree α. According to Lemma 2, system (13) can achieve stable in finite time. Then, by Lemma 3, one can get the conclusion that coupled system (13) could achieve finite time stable. That is, coupled system (1) under controller (14) can achieve finite time consensus. □

7. Simulation

In this simulation section, some examples are introduced to illustrate the theories. Example 1 is provided to illustrate the bounded consensus, while Example 2 presents the finite time consensus of variable quantity agent set.

Example 1. We introduced this example to illustrate the consensus of varying number multiple agents in Theorem 1 and Theorem 2. We choose $H_{ij} = x_j$, $i, j \in V$. It is assume 8 agents in the network interacting over the directed topology G. We assume the agent $1, 2, 3, 4, 5$ is follower; agent $6, 7, 8$ is leader. We assume the quantity of leader agent set switches from 2 to 3 during the process; the quantity of follower agent set switches from 2 to 5 during the process, see Fig. 2.

The consensus can be achieved if we choose $k = 2.5$, which is present in Fig. 3 We choose the $s_i = 1$, $i = 1, 2$ and $s_i = -1$, $i = 3, 4, 5$. Let us define $\epsilon(x) = \sum_{i=1}^{N(t)} \|r_i x_i(t) - \sum_{j \in E} c_{ij} r_i x_j\|$, $\epsilon(v) = \sum_{i=1}^{N(t)} \|r_i v_i(t) - \sum_{j \in E} c_{ij} r_j x_j\|$. Then, the bounded errors between leader and follower are presented in Fig. 4.

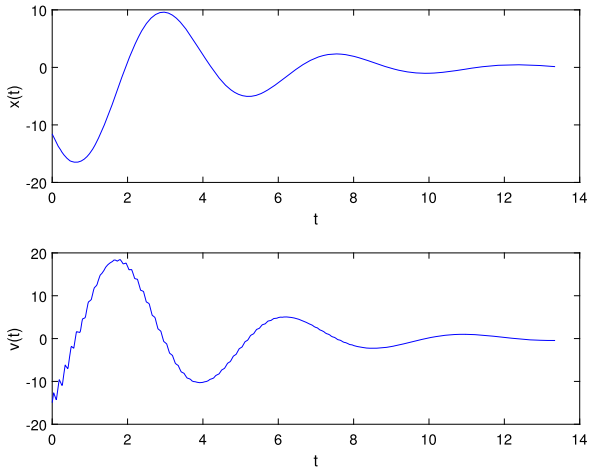


Fig. 4. Evolution trajectories of tracking errors ϵ, where $\epsilon(x) = \sum_{i=1}^{N(t)} \|r_i x_i(t) - \sum_{j\in E} c_{ij} r_i x_j\|$, $\epsilon(v) = \sum_{i=1}^{N(t)} \|r_i v_i(t) - \sum_{j\in E} c_{ij} r_j x_j\|$. The scaled consensus of coupled systems.

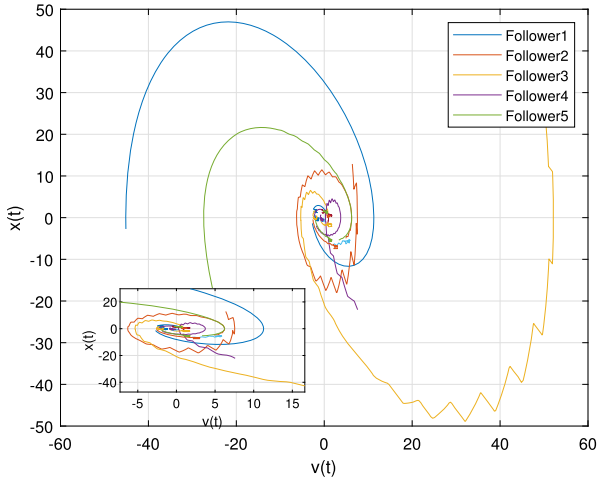


Fig. 5. The positions of follower agents achieve bounded consensus in finite time with $\alpha = 0.6$.

Example 2. Consider the same dynamics of coupled system in Example 1. Under the controller protocol (14), the trajectories of system's position and velocity state are presented in Fig. 5 with $\alpha = 0.6$. We define the errors between leaders and follower as $\epsilon'(x) = \sum_{i=1}^{N(t)} \|r_i x_i(t) - \sum_{j\in E} c_{ij} r_i x_j\|$, $\epsilon'(v) = \sum_{i=1}^{N(t)} \|r_i v_i(t) - \sum_{j\in E} c_{ij} r_j x_j\|$. The evolution trajectories of errors can be found in Fig. 6.

8. Conclusion

In this paper, we consider the interval consensus of variable agents of second-order multi-agent systems with variable quantity agents, every agent drives the agents to a desired interval. We present two kinds of framework, which is according to the agents themselves or their neighbors' state imposes as the interval constraints. Then, the scaled consensus under the varying quantity of agents with interval constraints is considered. We have presented the interval consensus could be achieved if the communication topology graph has a spanning tree across the time interval and the damping gain is larger than the threshold. Finally, it is worth mentioning that the finite-time consensus is also considered in this paper. Then, the consensus problem under variable number of agent with related topology conditions will be considered in the future.

CRediT authorship contribution statement

Ku Du: Writing – review & editing. **Zhiming Li:** Review. **Yu Kang:** Supervision.

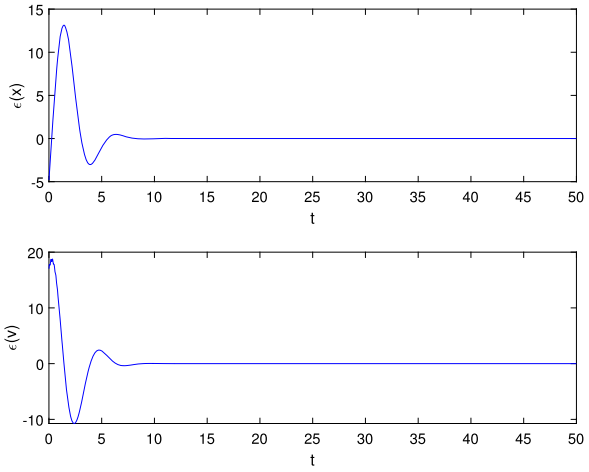


Fig. 6. Evolution trajectories of tracking errors ϵ, where $\epsilon'(x) = \sum_{i=1}^{N(t)} \|r_i x_i(t) - \sum_{j\in E} c_{ij} r_i x_j\|$, $\epsilon'(v) = \sum_{i=1}^{N(t)} \|r_i v_i(t) - \sum_{j\in E} c_{ij} r_j x_j\|$. The scaled consensus of coupled systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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